1. Minimizing lateness problem.

- Single resource processes one job at a time.
- Job i requires t_i units of processing time and is due at time d_i.
- If i starts at time s_i, it finishes at time $f_i = s_i + t_i$.
- Lateness: $\ell_i = \max\{0, f_i - d_i\}$.
- Goal: schedule all jobs to minimize maximum lateness $L = \max\{\ell_i \mid 1 \le i \le n\}$.

i	1	2	3	4	5	6	7
t_i	3	3	5	1	2	3	2
d_i	7	8	9	8	9	14	16

(1) 貪婪演算法, d_i 愈早, i 項愈早排

i	1	2	3	4	5	6	7
t_i	3	3	5	1	2	3	2
s_i	0	3	7	6	12	14	17
f_i	3	6	12	7	14	17	19
d_i	7 ①	8 ②	9 ④	8 ③	9 ⑤	14 ⑥	16 ⑦
l_i	0	0	3	0	5	3	3

5分 → $L = \max\{0, 3, 5\} = 5$

(2) d_i 由愈早，job 愈早執行

5分

如何

排程

Algorithm：

① d_i 由小排到大，t_i 根據 d_i 結果一起排

② $s[1] = 0$，$f[1] = 0$

assume there are N jobs.

③ for 1 to N

$$s[i] = f[i]$$

$$f[i] = s[i] + t[i]$$

找

lateness

$$l[i] = \max\{0, f[i] - d[i]\}$$

④ $L = {}^{*}\max\{l, l+n\}$